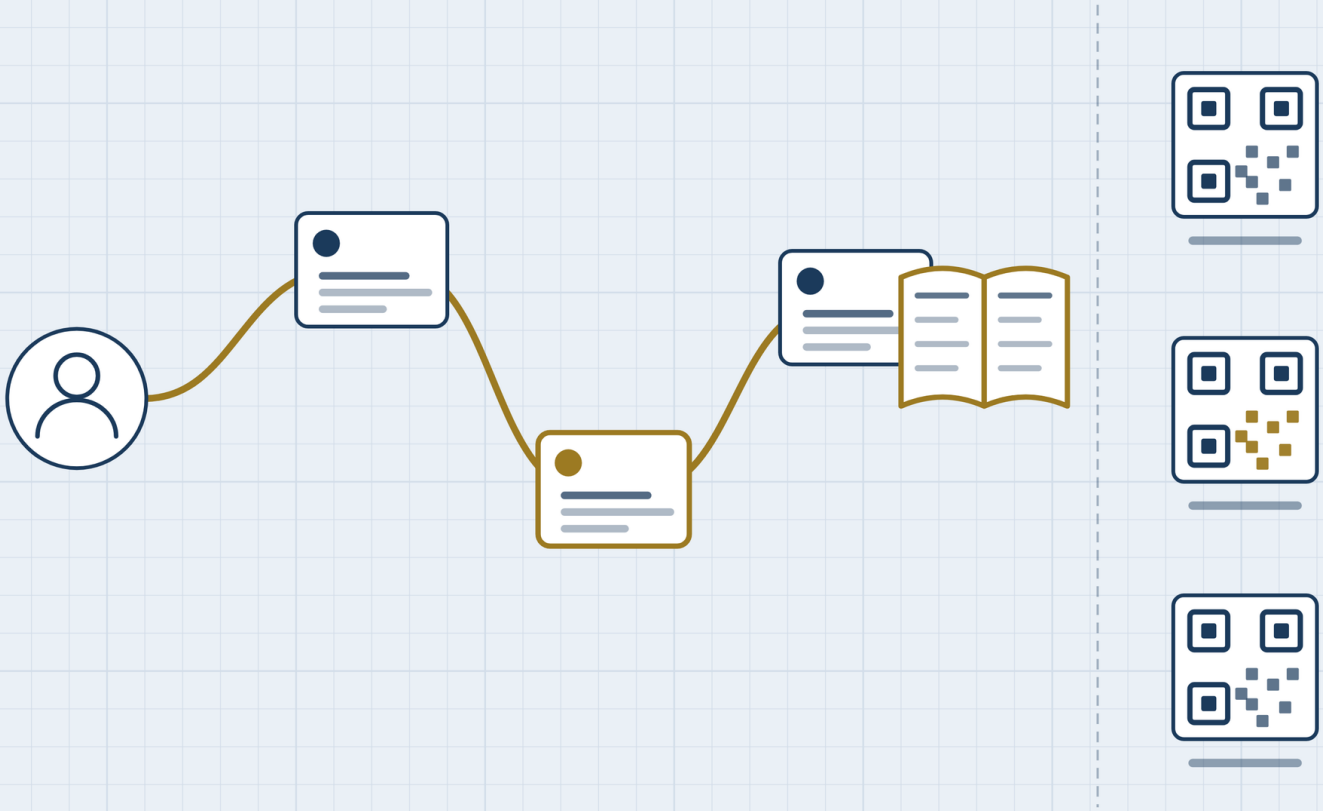


Designing an online course

A demo course that teaches how to build an effective online course — using the very features it showcases.

Mohamed EL AFRIT

Designing an online course

A demo course that teaches how to build an effective online course — using the very features it showcases.

Mohamed EL AFRIT

First edition

© 2026 Mohamed EL AFRIT — All rights reserved.

Students may use this course book for personal and educational purposes. Any other reproduction or distribution, in whole or in part, in any form or by any means, requires the prior written permission of the author.

All trademarks and product names mentioned in this book are the property of their respective owners.

The online course

Audio capsules, activities with instant feedback, flashcards, and the quiz:

https://example.org/cours

About the Author

This text introduces the author of the course: their background, what they teach, and what led them to write this course. In a course created from the template, the author writes and approves it in outils/livret.py; the course book then prints it word for word.

How to Use This Course Book

This course book contains the online course: its chapters, each opening with its contents, then a glossary and an index. You can work through it on its own, pencil in hand; the website picks up where paper leaves off, so you can listen, experiment, and practice.

The margin is yours



ONLINE
The online course

The wide outer margin has two uses: your notes — questions, rephrasings, sketches — and the QR codes. Each code opens, at the right spot, the online resource that extends the paragraph next to it: just point your phone's camera at it. On screen, click the code or its caption.

LISTEN	The section's audio capsule: the essentials, told in one to two minutes.
PODCAST	The chapter's podcast episode, to listen to before you read or to review.
GLOSSARY	A course term, defined in the margin; online, every place the course uses it.
ONLINE ACTIVITY	The exercise in the nearby box, with feedback as soon as you answer.
ANSWERS ONLINE	The answers to a set of questions, explained one by one.
ONLINE	A page of the site: the chapter, a resource, a page on another site.

The answers are online

No answers are printed. Work on paper first, then scan the activity's QR code: every answer is explained there, the correct one and the others alike.

When the text refers to another part of the book, it gives its page, like this: (p. 12).

The boxes

You can recognize each box by its icon, its label, and its border, even when it is printed in black and white.

KEY TAKEAWAYS
The essentials of a chapter, to know by heart.

KEY CONCEPT
A definition or a rule to keep in mind.

PITFALL
The most common mistake, and how to avoid it.

ACTIVITY
An exercise to do on paper, then to check online.

CHECKPOINT
A few questions to make sure you have the essentials down.

GOING FURTHER
Optional further reading.

READING THIS DIAGRAM, ELEMENT BY ELEMENT
A course diagram, explained element by element.

The course website

The website and this book mirror each other: the chapters have the same titles and the same sections. There, you can enlarge the diagrams, get feedback on the activities, and listen to the capsules.

Contents

How to Use This Course Book **4**

Chapter 1 · Scoping your course **7**

1. Why scope before producing 8
2. Who is learning, exactly? 8
3. Objectives you can verify 9
4. The through-line 9
5. A trade-off to settle 9
6. The chapter in one diagram 10
7. Review 11

Chapter 2 · Structuring the content **13**

1. From syllabus to sequences 14
2. Thinking in micro-learning 14
3. Building a progression 15
4. Balancing theory and practice 15
5. Review 15

Chapter 3 · Creating the resources **17**

1. Three formats, three intentions 18
2. Complementarity, not redundancy 18
3. Producing fast, without producing badly 18
4. Making resources accessible 19
5. Review 20

Glossary **21**

Index **23**

CHAPTER 1

Scoping your course

A badly scoped course is hard to repair: you notice on the third session, when it is too late to start over. This chapter sets out the three decisions to make before writing a single line. It also doubles as a showcase: every engine component is exercised here at least once.

Duration: 30 min

ONLINE
Chapter 1

LISTEN · 0 min 47 s
What this chapter asks you to decide

PODCAST
Episode 1 — Chapter 1

CHAPTER OBJECTIVES

- **Describe a target audience** by its prerequisites and constraints, not by its degree title.
- **Write a verifiable objective** using an observable action verb.
- **Choose a through-line** that runs across the course and holds it together.

IN THIS CHAPTER

1. Why scope before producing 8
2. Who is learning, exactly? 8
3. Objectives you can verify 9
4. The through-line 9
5. A trade-off to settle 9
6. The chapter in one diagram 10
7. Review 11

Key points 11

1. Why scope before producing

When you know your subject, the temptation is to open a document and start at the beginning. That is almost always a mistake: you then produce what you know, not what the learner needs.

Scoping means making three decisions, in order: **for whom**, **to do what**, and **around what**. Each one constrains the next. A defined audience narrows the possible objectives; verifiable objectives dictate the kind of activities; the through-line decides the order of chapters. Skipping one of the three does not save time — it moves the time to the rewrite.

GLOSSARY

Through-line
A continuous scenario or project that runs through the whole course and connects the chapters.

> **BEFORE YOU READ**
>
> *Choose your answer, then compare it with the one on the site.*
>
> Before reading: which of these three decisions do you think comes first?
>
> ☐ **A** Choosing the through-line, to hold the course together.
>
> ☐ **B** Defining the target audience, with its prerequisites and constraints.
>
> ☐ **C** Writing the learning objectives.
>
> **Hint:** *Ask which of the three constrains the other two, rather than the reverse.*

ANSWER ONLINE
Before you read

2. Who is learning, exactly?

"Master's students" describes nobody. A useful audience profile holds three facts: the **prerequisites actually acquired**, the **time available**, and the **practical constraints**. The third is the one people forget, and often the decisive one: a learner following the course on a commute will not watch a twenty-minute video.

Write that profile down, at the top of the syllabus. It will settle every difficult decision later — and it will stop you writing for the ideal learner, who does not exist.

GLOSSARY

Prerequisites
The knowledge or skills assumed before starting.

> **CHECKPOINT**
>
> **Describing an audience**
>
> *One answer per question; then check your answers online.*
>
> 1. Which of these descriptions is actually useful for scoping a course?
>
> ☐ **A** Second-year master's students.
>
> ☐ **B** Career changers with no technical background, available two hours a week, mostly on a phone.
>
> ☐ **C** A motivated and curious audience.
>
> 2. An objective is verifiable when…
>
> ☐ **A** it covers an important notion of the syllabus.
>
> ☐ **B** you can describe the exercise that would prove it reached.
>
> ☐ **C** it is written in a single sentence.

ANSWERS ONLINE
Checkpoint

3. Objectives you can verify

A learning objective states what the learner will be able to **do**, not what they will have "seen" or "understood". The difference is not cosmetic: it decides whether you can assess at all. "Understand the notion of an objective" cannot be observed. "Write an objective containing an action verb" can be observed, corrected and graded.

The test is simple: if you cannot picture the exercise that would prove the objective reached, it is badly written.

GLOSSARY

Learning objective
What the learner should be able to DO at the end of a sequence, written with an observable action verb.

ONLINE ACTIVITY

ACTIVITY

Match each item on the left with a letter on the right. Match each wording with the judgement that fits.

Objective wording	Judgement
1. Understand the importance of scoping.	A. Not verifiable: "know" does not say what to do.
2. Write an objective containing an action verb.	B. Verifiable: the task is observable and correctable.
3. Know the three scoping decisions.	C. Not verifiable: "understand" cannot be observed.
4. Rank three objectives by verifiability.	D. Verifiable: the proving exercise is obvious.

4. The through-line

A through-line is a case, a project or a scenario that runs across the whole course. It answers in advance the question the learner asks at every chapter: "what will this be used for?". This demo course has one — you are building a course while reading a course about building one.

ONLINE ACTIVITY
Spot the errors

SPOT THE ERRORS

Check the passages that contain an error; the correction is online. Here is the header of a syllabus written in a hurry. Three statements are a problem: spot them.

Syllabus — "Introduction to instructional design"

☐ 1. Audience: anyone interested in teaching.

☐ 2. Duration: six two-hour sessions, on site.

☐ 3. Objective: raise participants' awareness of teaching issues.

☐ 4. Assessment: a design report handed in at the last session.

☐ 5. Through-line: chapters will be illustrated with various examples.

5. A trade-off to settle

Scoping is not a theoretical exercise: it plays out in concrete decisions, usually taken under time pressure. Here is one of them.

ONLINE ACTIVITY
Two hours fewer than planned

ACTIVITY

Two hours fewer than planned

For each situation, choose what you would do; what happens next is online.

Your course was meant to run over six sessions. One has just been taken away. The content is written, and so are the activities.

You must cut the equivalent of two hours. What do you sacrifice?

- ☐ **A** I drop the activities and keep all the content: the syllabus will be covered.
- ☐ **B** I remove one whole chapter — the one that serves the through-line least.
- ☐ **C** I shorten every chapter by a third, evenly.

6. The chapter in one diagram

The three decisions and what each one constrains.



Diagram 1 — The three decisions and their order. Each narrows the field of the next: that is what makes scoping useful.

ONLINE
The interactive diagram

READING THIS DIAGRAM, ELEMENT BY ELEMENT

1. The audience — Prerequisites actually acquired, time available, practical constraints. The third is the forgotten one, and often the decisive one: a learner on a commute will not watch a twenty-minute video.

2. The objectives — What the learner will be able to DO. The test: if you cannot describe the exercise that would prove the objective reached, it is badly written — and you will not be able to assess.

3. The through-line — The case or project running across the whole course. It is also the arbiter on the day you must cut: what does not serve it is what you can do without.

7. Review

Flip the cards, then answer without looking at your notes.



REVIEW
The chapter's flashcards

KEY POINTS

- **Three decisions, in order:** for whom, to do what, around what. Each constrains the next.
- **An audience is described by its constraints**, not by its degree title.
- **An objective is verifiable or it is nothing:** if no exercise can prove it, it is badly written.
- **The through-line answers "what is this for"** before the learner asks.

CHAPTER 2

Structuring the content

Scoping settled for whom and to do what. What remains is deciding in which order, and in what pieces. This is where the difference is made between a course people finish and one they abandon at chapter three.

Duration: 25 min

ONLINE
Chapter 2

LISTEN · 0 min 47 s
Break up, without dismembering

CHAPTER OBJECTIVES

- **Break** a chapter into sequences that fit real attention spans.
- **Order** notions by what they assume known, not by their importance.
- **Balance** theory and practice without letting one crowd out the other.

IN THIS CHAPTER

1. From syllabus to sequences 14
2. Thinking in micro-learning 14
3. Building a progression 15
4. Balancing theory and practice 15
5. Review 15

Key points 11

1. From syllabus to sequences

A chapter is not a unit of learning: it is a unit of storage. What the learner actually goes through are **sequences** — blocks that start, end, and leave something behind.

A useful sequence holds three parts: a **question** that motivates it, **content** that answers it, and a **check** that proves it landed. Remove the question and the learner does not know why they are reading. Remove the check and nobody knows whether it worked — not even them.

ANSWERS ONLINE
Checkpoint — the sequence

CHECKPOINT

The sequence

One answer per question; then check your answers online.

1. What is most often missing from a course sequence?
   - ☐ A The question that motivates it.
   - ☐ B The check that proves it landed.
   - ☐ C The content itself.
2. A five-minute sequence counts as micro-learning when…
   - ☐ A it lasts under ten minutes.
   - ☐ B it stands on its own: a learner who stops there leaves with something usable.
   - ☐ C it can be read on a phone.

2. Thinking in micro-learning

Breaking up is not chopping up. A five-minute sequence that does not stand on its own is not micro-learning: it is a chapter cut in half. The test is simple — if the learner stops there, do they leave with something usable?

It is also a format decision. A short sequence can be read on a commute; a forty-minute one assumes a desk and a closed door. The audience profile written in chapter 1 settles this for you: re-read it rather than deciding on instinct.

ONLINE ACTIVITY

ACTIVITY

Match each item on the left with a letter on the right. Match each audience constraint with the format decision it imposes.

Audience constraint	Format decision
1. Reads mostly while commuting.	A. Long sequences acceptable, with a workshop closing the session.
2. Has two uninterrupted hours, once a week.	B. Short, self-contained sequences, readable on a phone.
3. Returns to the course after weeks away.	C. Explicit recap of what was learnt at the start of each chapter.
4. Has no technical prerequisites.	D. Every domain term defined at its first appearance.

3. Building a progression

The most common mistake is ordering notions by importance: starting with what matters most. But a learner can only take on a notion if they already hold the ones it assumes. The right order is that of **prerequisites**, not of importance.

A method that works: list your notions, then draw an arrow from each to those it assumes. Notions with no incoming arrow are your starting points. If you end up with a cycle — A assumes B which assumes A — then one of the two is badly cut.

SPOT THE ERRORS

Check the passages that contain an error; the correction is online. Here is a course outline. Three ordering decisions are a problem: spot them.

Outline — "Introduction to project management"

☐ 1. Chapter 1: the professional certifications in the field.

☐ 2. Chapter 2: vocabulary and roles in a project.

☐ 3. Chapter 3: a full case study on a real project.

☐ 4. Chapter 4: planning, estimating, tracking.

☐ 5. Chapter 5: introduction — why manage a project.

ONLINE ACTIVITY
Spot the errors

4. Balancing theory and practice

An entirely theoretical course produces learners who can recite. An entirely practical one produces learners who can repeat, and nothing more: as soon as the case changes, they are lost. The balance is not a matter of proportions but of alternation — a notion, its application, then the next notion.

The through-line acts as a guardrail here. If a sequence advances neither understanding nor the deliverable, it advances nothing at all — and it is the one to cut on the day a session is taken away from you.

5. Review

Flip the cards, then answer without looking at your notes.

KEY POINTS

- **A sequence has three parts:** a question, content, a check. One is always missing, and it is almost always the third.
- **Breaking up is not chopping up:** a short sequence must stand on its own.
- **The right order is that of prerequisites**, not of importance.
- **Alternate rather than apportion:** a notion, its application, the next one.

REVIEW
The chapter's flashcards

CHAPTER 3

Creating the resources

Text, audio, diagram: three formats, three different intentions. Confusing them is expensive — in production time, and in lost attention. This chapter says which to choose, and why duplication is almost always a mistake.

Duration: 25 min

ONLINE
Chapter 3

LISTEN · 0 min 50 s
Choose the format, not the tool

CHAPTER OBJECTIVES

- **Choose** the format from the intention, not from the tool at hand.
- **Make formats work together** instead of repeating each other.
- **Make every resource accessible** — which, done well, makes it better for everyone.

IN THIS CHAPTER

1. Three formats, three intentions 18
2. Complementarity, not redundancy 18
3. Producing fast, without producing badly 18
4. Making resources accessible 19
5. Review 20

Key points 11

1. Three formats, three intentions

Text is for what gets re-read: a definition, a procedure, an exact figure. **Audio** is for what carries better in a voice: an intention, a nuance, an encouragement. A **diagram** is for what no sentence can hold: a structure, a flow, a balance of forces.

The choice rests on the intention, not on the tool you happen to have. That is the common trap: you own a microphone, so you record; you own a diagram tool, so you diagram. A definition delivered by voice is lost — you would have to listen again, and nobody listens again.

ANSWER ONLINE
Before you read

BEFORE YOU READ

Choose your answer, then compare it with the one on the site.

Before reading: you must convey an exact retention period — fifteen minutes. Which format do you choose?

- ☐ **A** An audio capsule: a voice holds attention.
- ☐ **B** Text: an exact figure must be re-readable.
- ☐ **C** A diagram: a visual representation sticks better.

Hint: *Ask yourself what the learner will do with this figure in three weeks.*

2. Complementarity, not redundancy

A diagram that repeats the paragraph beside it is useless: it takes space and tires the reader. A good diagram shows what text cannot say — an order, a simultaneity, a proportion — while the text says what the diagram cannot show: the reason why.

The same rule holds for audio. A capsule that recites the page adds nothing for someone who has read it, and replaces nothing for someone who has not. A good capsule does something else: it situates, it warns of a trap, it says what really matters in what follows.

ONLINE ACTIVITY
One more capsule?

ACTIVITY

One more capsule?

For each situation, choose what you would do; what happens next is online.

Your chapter is written. It holds a clear diagram and two dense paragraphs on a difficult notion.

You have time to record a two-minute audio capsule. What do you have it say?

- ☐ **A** It summarises the chapter, for those who will not have time to read it.
- ☐ **B** It warns of the classic trap on this notion, and says what the difficulty will be good for.
- ☐ **C** It reads the two difficult paragraphs aloud.

3. Producing fast, without producing badly

Producing resources is where courses stall. Recording thirty capsules by hand is discouraging; generating them in bulk without checks produces takes nobody listens to before publication.

The workable path has two parts. First a **written script** for each resource, with its target duration: that is what you re-read and correct, not the recording. Then an **automatic check** before publication — duration, pace and silences for audio; contrast and alternative text for images. Whatever the check rejects goes aside, never online.

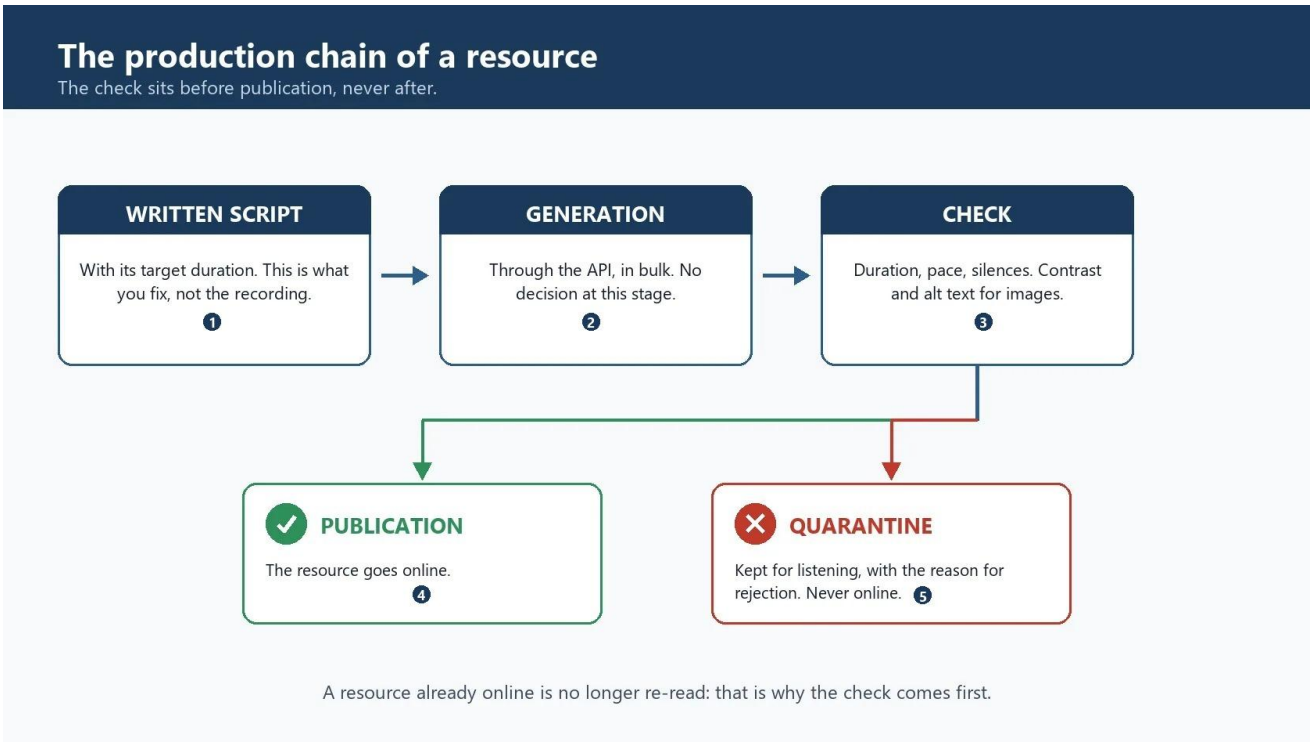


Diagram 1 — The production chain. The check sits before publication, never after: a resource already online is no longer re-read.

READING THIS DIAGRAM, ELEMENT BY ELEMENT

1. **The written script** — With its target duration. This is what you re-read and fix — never the recording. Fixing a take costs another generation; fixing a script costs a sentence.
2. **Generation** — In bulk, through the API. No decision is taken at this stage: everything was settled in the script.
3. **The check** — Duration, pace and silences for audio; contrast and alt text for images. It sits before publication, never after.
4. **Publication** — The resource goes online. It passed the check: you know what it is worth before anyone hears it.
5. **Quarantine** — Whatever the check rejects is kept for listening, with the reason — but never goes online. An inaudible take, once published, is only fixed by a student who complains.

ONLINE
The interactive diagram

4. Making resources accessible

An audio capsule without a transcript is unusable for a deaf person — and for anyone on a train without headphones. A diagram without alternative text does not exist for a screen reader. Insufficient contrast shows up first in bright sunlight, long before it becomes a matter of visual impairment.

GLOSSARY

Accessibility
The ability of content to be used by everyone, including people with disabilities.

REVIEW
The chapter's flashcards

This is the point people grasp last: accessibility is not done *for* a minority. It makes the resource usable in more situations, for everyone. A transcript serves the deaf person, the traveller, the one hunting for a specific word, and the search engine.

5. Review

Flip the cards, then answer without looking at your notes.

KEY POINTS

- **Format follows intention**, not the tool at hand. Text for what is re-read, audio for what is transmitted, diagrams for what cannot be said.
- **Duplication loses twice:** production time, and attention.
- **A written script, then an automatic check.** You fix the script, never the recording.
- **Accessibility is not done for a minority:** it makes the resource usable in more situations, for everyone.

Glossary

The course terms, as the site's glossary defines them, each one with its French equivalent and its definition.

The glossary online

A

Accessibility *accessibilité*
The ability of content to be used by everyone, including people with disabilities.

F

Flashcard *flashcard*
A two-sided active-recall card (question on one side, answer on the other).

Formative assessment *évaluation formative*
Assessment during learning, meant to give feedback rather than a grade.

H

Hook *accroche*
A short opening that motivates the learner by showing why the chapter matters.

L

Learning objective *objectif pédagogique*
What the learner should be able to DO at the end of a sequence, written with an observable action verb.

M

Microlearning *micro-apprentissage*
Breaking content into short, self-contained units.

Multiple-choice quiz *QCM*
A multiple-choice questionnaire; here graded by degree of certainty (CBM).

P

Prerequisites *prérequis*
The knowledge or skills assumed before starting.

S

Summative assessment *évaluation sommative*
A final assessment that measures and certifies learning.

Syllabus *syllabus*
The detailed course plan: chapters, objectives, durations and assessment.

T

Target audience *public cible*
The intended group of learners, with their prerequisites, level and constraints.

Through-line *fil rouge*
A continuous scenario or project that runs through the whole course and connects the chapters.

Index

The glossary terms, with the pages where the course uses them; in bold, the page where the margin defines each one.

A

Accessibility, **20**

L

Learning objective, **9**

P

Prerequisites, **8**, 15

S

Syllabus, 8

T

Through-line, **8**, 9, 11, 15